Kunta Sai Charan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.34
SRI CHAITHANYA JUNIOR COLLEGE	Class XII (Telangana State Board of Intermediate Education)	2023	91.10%
KKR GAUTHAM HIGH SCHOOL	Class X (Telangana State Board of Secondary Education)	2021	10.00%

EXPERIENCE

Equity Markets Analyst | Finlatics

- Analyzed equities using **technical indicators**, **candlestick patterns**, and valuation metrics (**P/E, P/B, EPS, ROE**).
- Assessed investment opportunities through **financial analysis**, **sector research**, and risk-return evaluation.
- Applied **DCF valuation**, **Excel-based financial modeling**, and **futures & options** concepts for risk hedging.

Research Intern | ECG Signal Digitization, IIT Hyderabad

- Built an **AI pipeline** that digitizes **printed ECG images** into signals doctors and researchers can analyze.
- Solved messy scan layouts using **Detectron2 Faster R-CNN** lead cropping and **Unet/ResNet** segmentation.
- Trained a custom **PyTorch** time-series model with **Huber Loss** and **Cosine Annealing** for accurate signals.

PROJECTS

Multi-Modal Commuter Route Planner

- Solved commuters' fast-vs-cheap dilemma by offering **Fastest, Cheapest, and Best-Value** TSRTC routes.
- Designed a **Best-Value score** balancing time, fare, and safety using **Dijkstra & A*** on weighted transit graphs.
- Ranked routes using **weather, safety, and ETA** signals so users can choose the best trip confidently.

Real-Time Stock Trading Platform

- Built a fast **C++ trading engine** that matches buy and sell orders quickly for a smooth user experience.
- Added **WebSocket** alerts so every client sees **live stock prices** in real time without refreshing pages.
- Designed a **parallel queue** with **MySQL** storage so orders stay reliable and data persists safely under load.

Digital Banking Simulator

- Built a **C++ banking app** for customers to manage accounts with **O(1) lookup** via **unordered_map**.
- Designed modular **OOP** Transaction and Bank classes so deposits, withdrawals, and transfers are easy to track per account.
- Added **per-account history** using **STL vectors** and unique auto-incrementing IDs so users can review activity.

Document Q&A RAG System

- Built a **production-oriented Google Drive RAG chatbot** for **semantic retrieval** and context-aware Q&A.
- Engineered **document chunking**, **Ollama embeddings**, and persistent **ChromaDB** vector search.
- Integrated **Qwen LLMs** for **source-grounded responses** through a modular **Python/FastAPI** pipeline.

SKILLS

Programming Languages: C, C++, Python, SQL

Data Science & ML: NumPy, Pandas, Matplotlib, Scikit-learn, OpenCV

LLM & RAG: RAG (Retrieval-Augmented Generation), Ollama, ChromaDB, Qwen LLMs, Vector Search

Developer Tools & Frameworks: Git, GitHub, FastAPI, Streamlit, Jupyter Notebook, VS Code, MySQL

RELEVANT COURSES

Mathematics: Probability & Statistics, Linear Algebra, Calculus, Differential Equations, Transformation Techniques

Computer Science: Data Structures & Algorithms, Object-Oriented Programming, Machine Learning, Computer Vision, Database Management Systems, Operating Systems

POSITIONS OF RESPONSIBILITY

PR Head | Milan 2025: Managed public relations, media outreach, and collaborations to strengthen Milan's campus presence.

Multimedia Head | Milan 2025: Led a 40-member creative team to deliver multimedia content and enhance Milan's digital brand.

ACHEIVMENTS

Elan & Vision 2024: Won 2nd prize for designing an **AI-powered e-waste management vending machine**.

Milan 2025: Gold medalist in Billiards.

EXTRACURRICULARS

Problem Solving: Enjoys identifying real-life inefficiencies and building tech-driven, automated solutions to simplify everyday problems.

Photography & Design: Pursues photography and design as a creative outlet, blending visual thinking with technical work.

Painting: Practices painting as a regular creative pursuit.

Reading: Reads regularly to build perspective and inform problem-solving approaches.